① Să se dezvolte în serie de puteri ale lui x, $f(x)=\arcsin x$ și să se determine suma seriei $\sum_{n\ge1}\frac{1^2\cdot3^2\cdot\ldots\cdot(2n-1)^2}{2^{2n}(2n+1)!}$

$$f'(x)=(\arcsin x)'=\frac{1}{\sqrt{1-x^2}}=(1-x^2)^{\frac12}=1+\sum_{n\ge1}\frac{(-x^2)^n}{n!}\left(-\frac12\right)\left(-\frac32\right)\cdots\left(-\frac{1-2n}{2}\right)$$

$$=1+\sum_{n\ge1}\frac{1\cdot3\cdot\ldots\cdot(2n-1)}{n!\cdot2^n}x^{2n}$$

$$f(x)=\int_0^x f'(y)\,dy=y\Big|_0^x+\sum_{n\ge1}\frac{1\cdot\ldots\cdot(2n-1)}{n!\cdot2^n}\cdot\frac{y^{2n+1}}{2n+1}\Big|_0^x=$$

$$f(x)=x+\sum_{n\ge1}\frac{1\cdot3\cdot\ldots\cdot(2n-1)}{n!\,2^n(2n+1)}\cdot x^{2n+1}=x+\sum_{n\ge1}\frac{(2n-1)!!}{(2n)!!}\cdot\frac{x^{2n+1}}{2n+1}$$

pt. $|x|<1$

$$f(x)=x+\sum_{n\ge1}\frac{1\cdot3\cdot\ldots\cdot(2n-1)}{2\cdot4\cdot6\cdot\ldots\cdot2n}\cdot\frac{x^{2n+1}}{2n+1}=x+\sum_{n\ge1}\frac{1^2\cdot3^2\cdot\ldots\cdot(2n-1)^2}{1\cdot2\cdot3\cdot4\cdot\ldots\cdot(2n-1)\cdot2n}\,\frac{x^{2n+1}}{2n+1}$$

$$f(x)=x+\sum_{n\ge1}\frac{1^2\cdot3^2\cdot\ldots\cdot(2n-1)^2}{(2n+1)!}x^{2n+1}\qquad |x|<1$$

$$f\left(\tfrac12\right)=\frac12+\sum_{n\ge1}\frac{1^2\cdot\ldots\cdot(2n-1)^2}{(2n+1)!\cdot2^{2n+1}}\ \Big|\cdot2\Rightarrow 2f\left(\tfrac12\right)=1+\sum_{n\ge1}\frac{1^2\cdot\ldots\cdot(2n-1)^2}{(2n+1)!\,2^{2n}}$$

$$\sum_{n\ge1}\frac{1^2\cdot\ldots\cdot(2n-1)^2}{(2n+1)!\,2^{2n}}=2f\left(\tfrac12\right)-1=2\cdot\frac{\pi}{6}-1=\frac{\pi}{3}-1$$

② Să se dezvolte în jurul punctului $x_0=0$ și $x_0=-2$, funcția $f(x)=\frac{x}{x^2-8x+15}$

$$f(x)=\frac{x}{x^2-8x+15}=\frac{x}{x^2-3x-5x+15}=\frac{x}{x(x-3)-5(x-3)}=\frac{x}{(x-3)(x-5)}$$

$$f(x)=\frac{A}{x-3}+\frac{B}{x-5}$$

$$Ax-5A+Bx-3B=x\Rightarrow\begin{cases}A+B=1\ |\cdot3\\-5A-3B=0\end{cases}\Rightarrow\begin{matrix}3A+3B=3\\-5A-3B=0\end{matrix}\Rightarrow\begin{matrix}-2A=3\\A=-\frac32\\B=\frac52\end{matrix}$$

Pt $x_0=0$

$$f(x)=-\frac32\cdot\frac{1}{x-3}+\frac52\cdot\frac{1}{x-5}$$

$$\frac{1}{x-3}=-\frac13\cdot\frac{1}{1-\frac{x}{3}}=-\frac13\sum_{n\ge0}\left(\frac{x}{3}\right)^n;\ \left|\frac{x}{3}\right|<1\Rightarrow|x|<3$$

$$\frac{1}{x-5}=-\frac15\cdot\frac{1}{1-\frac{x}{5}}=-\frac15\sum_{n\ge0}\left(\frac{x}{5}\right)^n;\ \left|\frac{x}{5}\right|<1\Rightarrow|x|<5$$

$$\Rightarrow f(x)=-\frac{\cancel{3}}{2}\cdot\left(-\frac{1}{\cancel{3}}\right)\cdot\sum_{n\ge0}\left(\frac{x}{3}\right)^n+\frac{\cancel{5}}{2}\cdot\left(-\frac{1}{\cancel{5}}\right)\sum_{n\ge0}\left(\frac{x}{5}\right)^n$$

$$f(x)=\frac12\sum_{n\ge0}x^n\left(\frac{1}{3^n}-\frac{1}{5^n}\right)\qquad|x|<3$$

$$x=3\Rightarrow f(3)=\frac12\sum_{n\ge0}3^n\left(\frac{1}{3^n}-\frac{1}{5^n}\right)=\frac12\sum_{n\ge0}\left[1-\left(\frac35\right)^n\right]$$

$\Rightarrow\sum$ e divergentă

$$x=-3\Rightarrow f(-3)=\frac12\sum_{n\ge0}(-3)^n\left(\frac{1}{3^n}-\frac{1}{5^n}\right)=\frac12\sum_{n\ge0}\left[\left(\frac35\right)^n-1\right]$$ neconverg. către 0

$\Rightarrow\sum$ divergentă

Pt $x_0=-2$

$$\frac{1}{x-3}=\frac{1}{(x+2)-5}=-\frac15\,\frac{1}{1-\frac{x+2}{5}}=-\frac15\sum_{n\ge0}\left(\frac{x+2}{5}\right)^n,\ \left|\frac{x+2}{5}\right|<1$$

$$\frac{1}{x-5}=\frac{1}{(x+2)-7}=-\frac17\,\frac{1}{1-\frac{x+2}{7}}=-\frac17\sum_{n\ge0}\left(\frac{x+2}{7}\right)^n,\ \left|\frac{x+2}{7}\right|<1$$

$$f(x)=-\frac32\cdot\left(-\frac15\right)\cdot\sum_{n\ge0}\left(\frac{x+2}{5}\right)^n+\frac52\cdot\left(-\frac17\right)\sum_{n\ge0}\left(\frac{x+2}{7}\right)^n$$

$$f(x)=\frac{3}{10}\sum_{n\ge0}\left(\frac{x+2}{5}\right)^n-\frac{5}{14}\sum_{n\ge0}\left(\frac{x+2}{7}\right)^n=\sum_{n\ge0}(x+2)^n\left(\frac32\cdot\frac{1}{5^{n+1}}-\frac52\cdot\frac{1}{7^{n+1}}\right)$$

$|x+2|<5$

$$x+2=5\Rightarrow\sum_{n\ge0}\left(\frac{3}{10}-\frac{5^{n+1}}{2\cdot7^{n+1}}\right)$$ divergentă pt că nu tind la 0

$$x+2=-5\Rightarrow\sum_{n\ge0}(-1)^n\left[\frac{3}{10}-\frac{5^{n+1}}{2\cdot7^{n+1}}\right]$$ diverg. pt. că nu tind la 0

③ Fie $\sum_{n\ge1}x_n=\sum_{n\ge1}(-1)^{n-1}\cdot\frac{3^n}{n}\cdot x^{2n}$ Determinați mult. de converg. și suma seriei. Arătați că $\sum_{n\ge1}(-1)^{n-1}\cdot\frac{1}{n(2n+1)}=\ln2+\frac{\pi}{2}-2$

fie $(-1)^{n-1}\cdot\frac{3^n}{n}=a_n$

Notăm $y=x^2$ astfel $\sum_{n\ge1}x_n=\sum_{n\ge1}(-1)^{n-1}\cdot\frac{3^n}{n}\cdot y^n$

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right|=\lim_{n\to\infty}\left|\frac{(-1)^n\cdot\cancel{3^{n+1}}\,3}{n+1}\cdot\frac{n}{(-1)^{n-1}\cdot\cancel{3^n}}\right|=\lim_{n\to\infty}3\cdot\frac{n}{n+1}=3$$

$$R_y=\frac13\Rightarrow I_{c_y}=\left(0;\frac13\right)\Rightarrow I_{c_x}=\left(-\frac{1}{\sqrt3};\frac{1}{\sqrt3}\right)$$

$$x=\pm\frac{1}{\sqrt3}\Rightarrow\sum_{n\ge1}x_n=\sum_{n\ge1}(-1)^{n-1}\cdot\frac{3^n}{n}\cdot\left(\frac{\pm1}{\sqrt3}\right)^{2n}$$ conform Leibniz e convergentă

$$\Rightarrow M_c=\left[-\frac{1}{\sqrt3};\frac{1}{\sqrt3}\right]$$

fie $S:\left[-\frac{1}{\sqrt3};\frac{1}{\sqrt3}\right]\to\mathbb{R}$ $\quad S(x)=\sum_{n\ge1}(-1)^{n-1}\cdot\frac{3^n}{n}\cdot x^{2n}$

Notăm $t=3x^2\Rightarrow f(t)=\sum_{n\ge1}(-1)^{n-1}\cdot\frac{t^n}{n}$

$$f'(t)=\sum_{n\ge1}(-1)^{n-1}\cdot\frac{\cancel{n}t^{n-1}}{\cancel{n}}=\sum_{n\ge1}(-1)^n\cdot t^n=\frac{1}{1+t}\xRightarrow{\int_0^t}f(t)=\ln(t+1)$$

$$S(x)=f(3x^2)=\ln(3x^2+1)=\sum_{n\ge1}(-1)^{n-1}\frac{3^n\cdot x^{2n}}{n}$$

$$\int_0^{\frac{1}{\sqrt3}}\ln(3x^2+1)\,dx=\int_0^{1/\sqrt3}\sum_{n\ge1}(-1)^{n-1}\cdot\frac{3^n}{n}\cdot x^{2n}dx$$

$$\int_0^{1/\sqrt3}\sum_{n\ge1}(-1)^{n-1}\cdot\frac{3^n}{n}\cdot x^{2n}dx=\sum_{n\ge1}(-1)^{n-1}\cdot\frac{3^n}{n}\cdot\frac{x^{2n+1}}{2n+1}\Big|_0^{1/\sqrt3}=\sum_{n\ge1}(-1)^{n-1}\cdot\frac{3^n}{n}\cdot\frac{1}{(\sqrt3)^{2n+1}\cdot(2n+1)}$$

$$=\sum_{n\ge1}\frac{(-1)^{n-1}}{n(2n+1)}\cdot\frac{1}{\sqrt3}\ (I)$$

$$\int_0^{1/\sqrt3}\ln(3x^2+1)dx=\int_0^{1/\sqrt3}x'\ln(3x^2+1)dx=x\ln(3x^2+1)\Big|_0^{1/\sqrt3}-\int_0^{1/\sqrt3}x\cdot\frac{6x}{3x^2+1}dx=$$

$$=\frac{1}{\sqrt3}\ln\left(3\cdot\frac13+1\right)-\int_0^{1/\sqrt3}\frac{6x^2}{3x^2+1}dx=\frac{1}{\sqrt3}\ln2-2\int_0^{1/\sqrt3}1-\frac{1}{3\left(x^2+\frac13\right)}dx$$

$$=\frac{\ln2}{\sqrt3}-2\left[x-\frac{\sqrt3}{3}\operatorname{arctg}(x\sqrt3)\right]\Big|_0^{1/\sqrt3}=\frac{\ln2}{\sqrt3}-2\left[\frac{1}{\sqrt3}-\frac{1}{\sqrt3}\operatorname{arctg}1\right]$$

$$=\frac{\ln2}{\sqrt3}-2\left[\frac{1}{\sqrt3}-\frac{1}{\sqrt3}\cdot\frac{\pi}{4}\right]=\frac{\ln2}{\sqrt3}-\frac{2}{\sqrt3}+\frac{\pi}{2\sqrt3}\ (II)$$

$$(I);(II)\Rightarrow\frac{1}{\sqrt3}\left(\ln2-2+\frac{\pi}{2}\right)=\sum_{n\ge1}\frac{(-1)^{n-1}}{n(2n+1)}\cdot\frac{1}{\sqrt3}\Rightarrow\sum_{n\ge1}\frac{(-1)^{n-1}}{n(2n+1)}=\ln2-2+\frac{\pi}{2}$$

④ Să se determine cu ajutorul seriilor de puteri, sumele seriilor numerice:

a) $\sum_{n\ge2}\frac{(-1)^n\cdot(n+1)}{n!\cdot3^{2n+1}}$

$$e^x=\sum_{n\ge0}\frac{1}{n!}\cdot x^n\Rightarrow xe^x=\sum_{n\ge0}\frac{1}{n!}x^{n+1}\xRightarrow{\text{deriv.}}e^x+xe^x=\sum_{n\ge0}\frac{(n+1)}{n!}x^n$$

$\forall x\in\mathbb{R}$

$$x=-\frac19\Rightarrow e^{-1/9}\left[1+\left(-\frac19\right)\right]=\sum_{n\ge0}\frac{(n+1)}{n!}\cdot\left(-\frac19\right)^n\Rightarrow\frac{8}{9\sqrt[9]{e}}=\sum_{n\ge0}\frac{(-1)^n(n+1)}{n!\cdot3^{2n}}\ \Big|\cdot\frac13$$

$$\frac13\cdot\frac{8}{9\sqrt[9]{e}}=\sum_{n\ge0}\frac{(-1)^n(n+1)}{n!\cdot3^{2n+1}}=\frac{1\cdot1}{0!\cdot3}+\frac{-1\cdot2}{1!\cdot3^3}+\underbrace{\sum_{n\ge2}\frac{(-1)^n(n+1)}{n!\cdot3^{2n+1}}}_{S}$$

$$S=\frac{8}{27}\cdot e^{-\frac19}-\frac13+\frac{2}{27}$$

b) $\sum_{n\ge1}\frac{(-1)^n}{5^{2n}(2n+1)(2n+2)}$

$$\frac{1}{1+x}=\sum_{n\ge0}(-1)^n\cdot x^n\Rightarrow\frac{1}{1+x^2}=\sum_{n\ge0}(-1)^n\cdot x^{2n}\xRightarrow{\int_0^x}\operatorname{arctg}x=\sum_{n\ge0}(-1)^n\cdot\frac{x^{2n+1}}{2n+1}$$

$$\xRightarrow{\int_0^x}\int_0^x\operatorname{arctg}y\,dy=\sum_{n\ge0}(-1)^n\cdot\frac{1}{2n+1}\cdot\frac{x^{2n+2}}{2n+2}$$

$\|$

$$\int_0^x y'\operatorname{arctg}y\,dy=y\operatorname{arctg}y\Big|_0^x-\int_0^x y\cdot\frac{1}{1+y^2}dy=x\operatorname{arctg}x-\frac12\ln(1+y^2)\Big|_0^x$$

$$z=y^2\longrightarrow\frac12\,\frac{1}{1+z}dz$$

$$dz=2y\,dy$$

$$x\operatorname{arctg}x-\frac12\ln(1+x^2)=\sum_{n\ge0}(-1)^n\cdot\frac{1}{2n+1}\cdot\frac{x^{2n+2}}{2n+2},\ |x|<1$$

$$x=\frac15\Rightarrow\frac15\operatorname{arctg}\frac15-\frac12\ln\left(1+\frac{1}{25}\right)=\sum_{n\ge0}\frac{(-1)^n}{(2n+1)(2n+2)\cdot5^{2n+2}}\ \Big|\cdot5^2$$

$$5\operatorname{arctg}\frac15-\frac{25}{2}\ln\left(\frac{26}{25}\right)=\sum_{n\ge0}\frac{(-1)^n}{(2n+1)(2n+2)5^{2n}}$$

$$5\operatorname{arctg}\frac15-\frac{25}{2}\ln\left(\frac{26}{25}\right)=\frac{1}{1\cdot2\cdot1}+\sum_{n\ge1}\frac{(-1)^n}{(2n+1)(2n+2)5^{2n}}$$

$$\Rightarrow\sum_{n\ge1}\frac{(-1)^n}{(2n+1)(2n+2)5^{2n}}=5\operatorname{arctg}\frac15-\frac{25}{2}\ln\left(\frac{26}{25}\right)-\frac12$$

⑤ $f(x,y,z)=\frac{x}{\sqrt{x^2+y^2+z^2}}$ $\quad\Delta f=?$ (laplacianul fc. f)

$$\Delta f=\frac{\partial^2f}{\partial x^2}+\frac{\partial^2f}{\partial y^2}+\frac{\partial^2f}{\partial z^2}$$

$$\frac{\partial f}{\partial x}=\frac{\sqrt{x^2+y^2+z^2}-x\cdot\frac{\cancel{2}x}{\cancel{2}\sqrt{x^2+y^2+z^2}}}{x^2+y^2+z^2}=\frac{x^2+y^2+z^2-x^2}{(x^2+y^2+z^2)\sqrt{x^2+y^2+z^2}}=\frac{y^2+z^2}{(x^2+y^2+z^2)^{3/2}}$$

$$\frac{\partial^2f}{\partial x}=\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right)=\frac{-(y^2+z^2)\cdot\frac{3}{\cancel{2}}(x^2+y^2+z^2)^{1/2}\cdot(\cancel{2}x)}{(x^2+y^2+z^2)^3}=-\frac{3x(y^2+z^2)}{(x^2+y^2+z^2)^{5/2}}$$

$$\frac{\partial f}{\partial y}=\frac{-x\cdot\frac{\cancel{2}y}{\cancel{2}\sqrt{x^2+y^2+z^2}}}{x^2+y^2+z^2}=\frac{-xy}{(x^2+y^2+z^2)^{3/2}}$$

$$\frac{\partial^2f}{\partial y}=\frac{-x\cdot(x^2+y^2+z^2)^{3/2}+xy\cdot\frac{3}{\cancel{2}}(x^2+y^2+z^2)^{1/2}\cdot\cancel{2}y}{(x^2+y^2+z^2)^3}=\frac{-x(x^2+y^2+z^2)+3xy^2}{(x^2+y^2+z^2)^{5/2}}$$

$$=\frac{-x(x^2+y^2+z^2-3y^2)}{(x^2+y^2+z^2)^{5/2}}=\frac{-x(x^2+z^2-2y^2)}{(x^2+y^2+z^2)^{5/2}}$$

$$\frac{\partial^2f}{\partial z}=\frac{-x(x^2+y^2-2z^2)}{(x^2+y^2+z^2)^{5/2}}$$

$$\Delta f=\frac{-3x(y^2+z^2)-x(x^2+z^2-2y^2)-x(x^2+y^2-2z^2)}{(x^2+y^2+z^2)^{5/2}}=\frac{-x(3y^2+3z^2+x^2+z^2-2y^2+x^2+y^2-2z^2)}{(x^2+y^2+z^2)^{5/2}}$$

$$=\frac{-x(2y^2+2z^2+2x^2)}{(x^2+y^2+z^2)^{5/2}}=\frac{-2x}{(x^2+y^2+z^2)^{3/2}}$$